Section	Text
Main synthesis	The classified pathways show that repairability alone is not enough to guarantee product-life extension. In several pathways, the expected benefit depends on contextual conditions such as component access, spare-part availability, installation method, transport distance, service infrastructure, dimensional standardisation and client decision-making. This supports the project's central argument that circular window strategies need to be assessed as product and system conditions together.
Strongest pathway	The strongest pathway is P1 because both expert interviews and the component map support the same logic. Short-life components may be technically repairable, but repair can fail in practice if access is blocked by finishes, reveals, installation detailing or the wall and window interface. This is best classified as an implementation gap rather than a direct rebound effect.
Most rebound-relevant pathway	P3 is the most rebound-relevant pathway because easier component replacement may encourage more frequent cosmetic or performance upgrading before functional failure. However, this should be presented cautiously as a potential behavioural rebound pathway, not as a measured rebound effect.
Most lifecycle-relevant pathway	P4 is the most lifecycle-relevant pathway because modularity or high performance may introduce additional material, connectors or technical complexity. This does not mean modularity is bad, but it means the expected benefit must be verified through lifecycle scenario comparison.
Most design-relevant pathway	P5 is especially relevant for architects because retaining a long-life frame can avoid material replacement, but it can also restrict future glazing upgrades or performance improvement. This shows why long service life must be paired with future upgrade compatibility.
Most system-relevant pathway	P7 shows that direct reuse is not only a technical question. Even if a window can be removed, reuse may fail because of custom dimensions, missing documentation, certification uncertainty, transport and lack of local reuse infrastructure.
How to describe claim level	These pathways identify plausible mechanisms, not measured market-level effects. They are based on the rapid literature review, the VELFAC case, exploratory interviews and lifecycle scenario assumptions. They should be used as analytical pathways for screening and discussion, not as proof of prevalence or causal magnitude.
How this leads to Step 14	The pathways show what the preliminary framework must check. It must ask whether the function is preserved, which parts fail first, whether short-life parts can be replaced independently, whether repair is practically enabled, which actors must support the strategy, what trade-offs are introduced, whether rebound or displacement could occur, and whether the expected lifecycle benefit remains plausible.